

04 NEW ENERGY

In a recent blog post, Google's senior VP, Technical Infrastructure wrote about Google being set to reach 100% renewable energy in 2017. So, every single Google search done, every email you send via Gmail, every YouTube video you watch, every Google AdSense ad you ignore on sites, app you download from the Google Play Store... everything Google that you use uses up energy from a server somewhere. Heck, those of you using

electricity producer (such as a wind farm or a solar farm).

Hydrogen is another fuel that's showing promise, and it's relatively cheap to make because it's extracted from water, and when it is burnt, the only exhaust product is water, making it a very desirable fuel source indeed. Another way to use hydrogen power is to use hydrogen fuel cells which uses hydrogen and oxygen

PUT THESE ON

<http://dgit.in/SmrtPrkw>

<http://dgit.in/MITwear>

Thankfully, we're not predicting that smartwatches will do anything spectacular in 2017. However, we do

think that the wearable space is set to take off in other ways. Let us explain that.

Take for instance the MagicBands used in Disney's theme parks. These

are bands that you wear on your wrist that contain information about you, what you've paid for, even what you want to have for lunch (if you choose to enter that data). What does this result in? No waiting in ticket lines, no need to worry about hotel room keys (the band is the key), and if you ordered your lunch, no waiting, it is prepared and kept ready for you, just scan the band and take your lunch. You update your selections at the many kiosks available at Disney World, by using their app, or on the MagicBand website. Not only does this make the consumer's life a lot easier, by tracking magic bands, the entertainment park is able to make sure there are enough staff to assist at locations where there is an extra crowd, for example, and also it cuts down on costs of running the place. This isn't new, or even unique to Disney parks, but we see it taking off elsewhere.



CORADIA ILINT, REGIONAL TRAIN, SALZGITTER (GERMANY).
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This train actually runs on hydrogen!

Google's free DNS servers... every time you click on a domain name you haven't been to before, you ask a Google server for an IP address... All of that humongous data flow, all of a million or so servers (Google's datacentre costs are supposedly \$20 billion a year), all of the energy used by Google offices... will be powered by renewable energy!

Now that's quite a feat, and Google will be the first large company to achieve this. Google is already the world's largest corporate buyer of renewable energy (non-fossil fuel energy), and it only buys solar and wind energy currently. Understand that Google is not off the grid, running on its own solar panels and windmills... their servers are way too important to risk a cloudy day suddenly resulting in YouTube going down! Instead, Google is paying the electric suppliers the premium rate necessary for the supplier to purchase that amount of power from a green and renewable

as the anode and cathode (respectively) of a battery to produce electricity and water as the only waste product.

Amongst others, Toyota is one car manufacturer working on a hydrogen powered car. We mention them because at the time of writing this their concept happens to be the one most recently released.

There's also a train that has been developed recently that will use hydrogen fuel cells to power it. It will run at speeds of up to 140 kmph, and carry as many as 300 passengers, with a range of between 600 to 800 km on a full tank of fuel!

05 WEARABLES

The minute we use this word, you will be thinking "Not more smartwatches please!". For good reason too,

because despite all the hype, smartwatches seem to have flopped royally. Apple somehow expects to make new records selling their Watch, but we're still skeptical.

MORE SHOCKS?

<http://dgit.in/REnergy100>

<http://dgit.in/ToyotaD>

<http://dgit.in/iLint>



SOURCE: FLICKR

Nano-printable circuits from MIT

An interesting development in wearables comes from MIT (again), where engineers have come up with a way to print really tiny circuits onto flexible surfaces. Unlike regular inkjet printing that was being tried before, the new method uses stamps to print pre-decided circuits onto a surface. While this means that you're not going to be using your home printer for this purpose, it does make for some really cool and tiny circuits, and that means the only limit is your imagination.